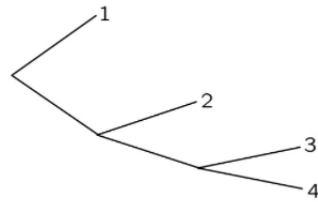


Homework 1- Due Date Feb 09, 11:59 P.M.

Problem 1. Sarah checks the fridge to find a snack. If she finds her favorite snack, she will stay home. If she does not find her favorite snack, then She will either stay home and order online or she will go grocery shopping. If she goes for grocery shopping, she might either take her dog with her or leave her dog at home. In the diagram below which one shows that Sarah orders online.



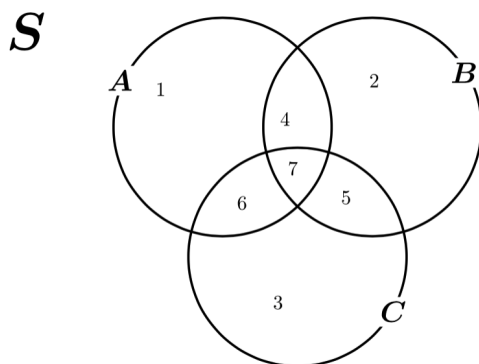
- (a) 1
- (b) 3
- (c) 2
- (d) 4

Problem 2. Suppose an experiment consists of tossing a two-sided coin four times. Consider the event $E = \{TTTT, HTTT, THTT, TTHT, TTTH\}$. Which of the followings describe E :

- (a) “At most once the coin turns heads”
- (b) “number of times tail occurs is more than two”
- (c) “At least three times the coin turns tails”
- (d) “number of tails is more than number of heads”

You may have more than one correct answer. Select every correct answer.

Problem 3. Sample space S as shown in the following figure is given. Which options are equivalent to the event $\{4, 6\}$. (Hint: Recall the set A minus B can be defined as $A - B = A \cap B'$)



- (a) $(B \cup C) \cap A$
- (b) $((B - C) \cup (C - B)) \cap A$
- (d) $(B \cap C' \cap A) \cup (B' \cap C \cap A)$

You may have more than one correct answer. Select every correct answer.

Problem 4. Let A and B be events on the same sample space, with $P(A) = 0.7$ and $P(B) = 0.5$. Can these two events be disjoint? Explain your answer.

Problem 5. Let A, B and C be pairwise disjoint subsets of the sample space. For each one of the following statements, determine whether it is true or false. Note: “False” means “It is not always true.”

- (a) $P(A^c) + P(B) \leq 1$
- (b) $P(A) + P(B) \leq 1$
- (c) $P(A \cup B \cup C) \geq P(A \cup C)$

Problem 6. A certain system can experience three different types of defects. Let A_i ($i= 1,2,3$) denote the event that the system has a defect of type i . Suppose that
 $P(A_1) = 0.12$, $P(A_2) = 0.07$, $P(A_3) = 0.05$
 $P(A_1 \cup A_2) = 0.13$, $P(A_1 \cup A_3) = 0.14$, $P(A_2 \cup A_3) = 0.10$,
 $P(A_1 \cap A_2 \cap A_3) = 0.01$.

- a. What is the probability that the system does not have a type 2 defect?
- b. What is the probability that the system has both type 2 and type 3 defects?

- c. What is the probability that the system has both type 2 and type 3 defects but not a type 1 defect?
- d. What is the probability that the system has at most two of these defects?

Problem 7. Let the sample space be the set of positive integers and suppose that $p(n) = \frac{2}{3^n}$, for $n = 1, 2, 3, \dots$. Find the probability of the set $\{5, 10, 15, \dots\}$, that is, the set of positive integers that are multiples of 5.

Problem 8. Out of the people in a seminar, 60% are mathematicians, 70% love coffee, and 48% fall into both categories. Determine the probability that a randomly selected person is neither a mathematicians nor a coffee lover.